

Linear Combination of Vectors

Let $V(F)$ be a vector space and $S = \{ \alpha_1, \alpha_2, \dots, \alpha_n \}$ be a finite subset of $V(F)$.

A vector $\alpha \in V$ is said to be a linear combination of the vectors $\alpha_1, \alpha_2, \dots, \alpha_n$ if α can be expressed as

$$\alpha = c_1 \alpha_1 + c_2 \alpha_2 + \dots + c_n \alpha_n \quad \text{--- (1)}$$

for some scalars $c_1, c_2, \dots, c_n \in F$.

Note:

* If we write α is l.c. of vectors $\alpha_1, \alpha_2, \dots, \alpha_n$, then we are to solve a system $AC = B$ of linear equations in unknowns

$$C = (c_1, c_2, \dots, c_n) \text{ and}$$

$$B = \alpha \text{ and}$$

the columns of the coefficient matrix A are α_k 's.

* If the system of linear equation $AC=B$ has no solution, then α can not be expressed as a linear combination of $\alpha_1, \alpha_2 \dots \alpha_n$.

1. Express $(2, 1, -6)$ as a linear combination of $(1, 1, 2)$, $(3, -1, 0)$ and $(2, 0, -1)$ in a real vector space $\mathbb{R}^3(\mathbb{R})$.

Solution: Let $\alpha = (2, 1, -6)$ and

$$\alpha_1 = (1, 1, 2), \alpha_2 = (3, -1, 0), \alpha_3 = (2, 0, -1)$$

Let $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$\alpha = c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3$$

$$(2, 1, -6) = c_1(1, 1, 2) + c_2(3, -1, 0) + c_3(2, 0, -1)$$

$$\Rightarrow c_1 + 3c_2 + 2c_3 = 2 \quad \text{--- (1)}$$

$$c_1 - c_2 + 0c_3 = 1 \quad \text{--- (2)}$$

$$2c_1 + 0c_2 - c_3 = -6 \quad \text{--- (3)}$$

Write augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 1 & -1 & 0 & 1 \\ 2 & 0 & -1 & -6 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 0 & -4 & -2 & -1 \\ 0 & -6 & -5 & -10 \end{array} \right]$$

$$R_3 \rightarrow 4R_3 - 6R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & 2 \\ 0 & -4 & -2 & -1 \\ 0 & 0 & -8 & -34 \end{array} \right]$$

Here Rank of A = rank of $[A|B]$ = no. of variables

so

$$-8C_3 = -34$$

$$C_3 = \frac{34}{8} = \frac{17}{4}$$

$$-4C_2 - 2C_3 = -1$$

$$\Rightarrow -4C_2 - 2\left(\frac{17}{4}\right) = -1$$

$$\Rightarrow -4C_2 - \frac{17}{2} = -1$$

$$\Rightarrow -4C_2 = \frac{17}{2} - 1$$

$$\Rightarrow -4C_2 = \frac{15}{2}$$

$$\Rightarrow C_2 = -\frac{15}{8}$$

$$C_1 + 3C_2 + 2C_3 = 2$$

$$C_1 + 3\left(-\frac{15}{8}\right) + 2\left(\frac{17}{4}\right) = 2$$

$$C_1 - \frac{45}{8} + \frac{34}{4} = 2$$

$$C_1 = 2 + \frac{45}{8} - \frac{34}{4}$$

$$= \frac{16 + 45 - 68}{8}$$

$$= -\frac{7}{8}$$

$$\therefore C_1 = -\frac{7}{8}, C_2 = -\frac{15}{8}, C_3 = \frac{17}{4}$$

$$\therefore (2, 1, 6) = -\frac{7}{8}(1, 1, 2) - \frac{15}{8}(3, -1, 0) + \frac{17}{4}(2, 0, -1)$$

$$= x =$$

2. In a vector space $\mathbb{R}^3(\mathbb{R})$, show that $(2, 5, 3)$ can not be expressed a linear combination of vectors $(1, -3, 2)$ $(2, -4, -1)$ and $(1, -5, 7)$

Solution:

Let $\alpha = (2, 5, 3)$ and

$$\alpha_1 = (1, -3, 2),$$

$$\alpha_2 = (2, -4, -1)$$

$$\alpha_3 = (1, -5, 7)$$

Let $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$\alpha = c_1 \alpha_1 + c_2 \alpha_2 + c_3 \alpha_3$$

$$(2, 5, 3) = c_1(1, -3, 2) + c_2(2, -4, -1) + c_3(1, -5, 7)$$

$$= (c_1 + 2c_2 + c_3, -3c_1 - 4c_2 - 5c_3, 2c_1 - c_2 + 7c_3)$$

$$c_1 + 2c_2 + c_3 = 2 \quad \text{--- (1)}$$

$$-3c_1 - 4c_2 - 5c_3 = 5 \quad \text{--- (2)}$$

$$2c_1 - c_2 + 7c_3 = 3 \quad \text{--- (3)}$$

augmented matrix of above system of equations

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ -3 & -4 & -5 & 5 \\ 2 & -1 & 7 & 3 \end{array} \right]$$

$$R_2 \rightarrow R_2 + 3R_1, \quad R_3 \rightarrow R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 11 \\ 0 & -5 & 5 & -1 \end{array} \right]$$

$$R_3 \rightarrow 2R_3 + 5R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 11 \\ 0 & 0 & 0 & 53 \end{array} \right]$$

hence

$$\text{rank of } A \neq \text{rank of } [A/B]$$

so above system has no solution

Hence α can not be expressed as a linear combination of the α_i 's.

3. Express $p(t) = 3t^2 + 5t - 5$ as a linear combination of the polynomial
 $p_1(t) = t^2 + 2t + 1$,
 $p_2(t) = 2t^2 + 5t + 4$
 $p_3(t) = t^2 + 3t + 6$

solution:

Given $p(t) = 3t^2 + 5t - 5$ and

$$p_1(t) = t^2 + 2t + 1$$

$$p_2(t) = 2t^2 + 5t + 4$$

$$p_3(t) = t^2 + 3t + 6.$$

Let $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$3t^2 + 5t - 5 = c_1 p_1(t) + c_2 p_2(t) + c_3 p_3(t)$$

$$= c_1 [t^2 + 2t + 1] + c_2 [2t^2 + 5t + 4] \\ + c_3 [t^2 + 3t + 6]$$

$$c_1 + 2c_2 + c_3 = 3 \quad \text{--- (1)}$$

$$2c_1 + 5c_2 + 3c_3 = 5 \quad \text{--- (2)}$$

$$c_1 + 4c_2 + 6c_3 = -5 \quad \text{--- (3)}$$

Solve above equations we get

$$C_1 = 3, C_2 = 1, C_3 = -2$$

$$\therefore p(t) = 3p_1 + p_2 - 2p_3$$

$$\underline{\underline{= x =}}$$

4. Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}$

and $x = \begin{bmatrix} 4 & 0 \\ 2 & 0 \end{bmatrix}$. Express X as a linear combination of A, B, C .

Solution:

Given $x = \begin{bmatrix} 4 & 0 \\ 2 & 0 \end{bmatrix}$ and

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, C = \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}$$

Let $p, q, r \in \mathbb{R}$ such that

$$X = pA + qB + rC$$

$$\begin{bmatrix} 4 & 0 \\ 2 & 0 \end{bmatrix} = p \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + q \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + r \begin{bmatrix} 0 & 2 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} p & p+2r \\ p+q & q-r \end{bmatrix}$$

here $p=4$,

$$p+2r=0$$

$$\Rightarrow 4+2r=0$$

$$\Rightarrow 2+r=0 \Rightarrow r=-2$$

$$p+q=2$$

$$\Rightarrow 4+q=2$$

$$q=-2$$

$$\therefore p=4, q=-2, r=-2$$

Hence the required relation is

$$X = 4A - 2B - 2C$$

5. If $\alpha = (3, 7)$ and $\beta = (2, 4)$, $\gamma = (-1, 1)$,
then the linear combination of α in terms
of β, γ is _____

$$= x =$$

6. In a vector space $\mathbb{R}^3(\mathbb{R})$, show that $(2, -5, 3)$ can not be expressed a linear combination of vectors $(1, -3, 2)$ $(2, -4, -1)$ and $(1, -5, 7)$